Low power & high throughput authenticated encryption chip for radio frequency applications

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ABSTRACT

A novel authenticated encryption chip for radio frequency communication application was developed. A range of low power techniques were applied to reduce the power consumption. The proposed authenticated encryption chip exhibits ultra low power dissipation of 139.2μW/MHz @1.8V V_{DD} and ultra high throughput of 1.07Gbps with a reasonable core area of 0.461mm^2 (46.8k gates) under SMIC 180nm technology.

Keywords: Authenticated Encryption, Radio Frequency, low power, chip design

1. INTRODUCTION

Radio Frequency (RF) communication is a technology that uses radio signals for information exchange. Due to the fast transmission speed and wireless, RF technology has now played an important role in Internet of Things (IoTs)[1], military security, and so on. However, as this technology widely applied, many security issues are increasingly prominent. Information leakage, property misappropriation and other issues occurred frequently, which restricted the development of RF communication.

Authenticated Encryption (AE) algorithm is a good way to ensure information security. It not only provides data encryption, but data source authentication as well, which can simultaneously guarantee the confidentiality, integrity and authentication of data. In addition, compared with the combination of encryption algorithm and message authentication code, AE algorithm can reduce the potential information leakage caused by data convergence between different algorithms effectively. The problem is that low area and low power are particularly important for hardware implementations.

The researches on AE Algorithm is Developing Worldwide. In order to solicit outstanding AE algorithms around the world, the Competition for Authenticated Encryption: Security, Applicability, and Robustness (CAESAR) came into being. After carefully assessing by cryptographers around the world, the candidate algorithms of the competition have been proved to have good security and reliability. Among all final round candidates, ASCON and ACORN were proved to be suitable for lightweight cases. AEGIS and OCB were designed for high-performance applications. Deoxys and COLM were good at defence in depth cases. A year after the final round of the CAESAR competition was announced, the National Institute of Standards and Technology (NIST) launched the Lightweight Cryptography Algorithms Competition. ASCON, as one of the candidate algorithms in the competition, became one of the winning algorithms in the final round in March 2021. After the evaluation and consideration from cryptographers worldwide, it is highly possible for the excellent algorithms to be promoted and applied as standards in industry.

In this article, we proposed a chip implementation of a lightweight AE algorithm ASCON in the RF field. It combines both encryption, decryption and authentication functions. The chip can adaptively adjust the encryption or decryption function according to the received data frame. It turned out to achieve a low core area of 0.461mm^2(1.4mm^2 with IOs) under SMIC 180nm technology with a considerable core power consumption of 46.4μW/MHz. At the operating frequency of 100MHz, our chip achieves a high throughput of 1.07Gbps.

2. THE AUTHENTICATED CIPHER OF ASCON

ASCON is a family of authenticated encryption and hashing algorithms designed for lightweight implementation. It has been selected as the primary choice for lightweight authenticated encryption in the final portfolio of the CAESAR competition (2014–2019)[2] and has just won the final round of the NIST Lightweight Cryptography competition (2019–2021)[3].

The mode of operation of ASCON is based on duplex sponge modes like MonkeyDuplex, but uses a stronger keyed initialization and keyed finalization function[4]. It contains four phases: *Initialization*, *Associated Data*, *Encryption/Decryption*, and *Finalization*, as figure 1 shows.

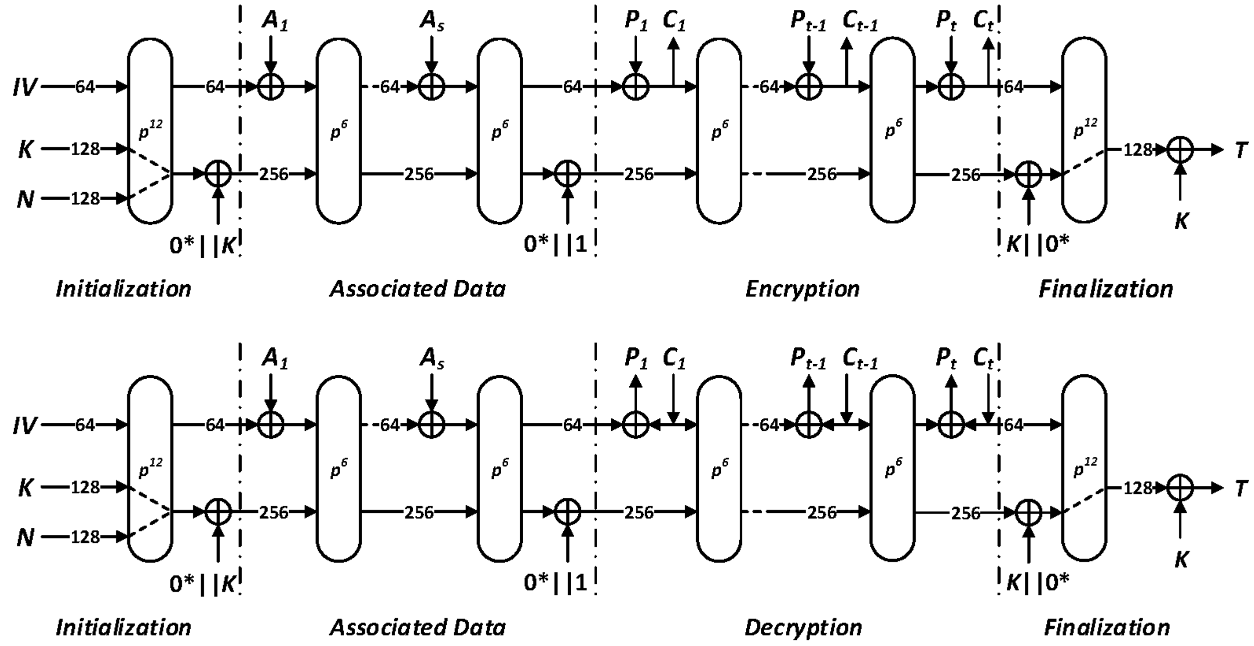


Figure 1. mode of operation of ASCON

The *Initialization* phase uses constant *IV* (64 bits), key *K* (128 bits) and nonce *N* (128 bits) to form 320-bit input state *S*. 12 rounds of the round transformation *p* are applied to the initial state *S*, followed by an XOR with the *K*. The *Associated Data* phase is optional. The associated data is padded into 64-bit data blocks. It doesn't need to be confidential but can't be modified by the attacker. Each padded associated data block is processed with state *S* by 6 rounds permutation unless no associated data is applied. After all these preparations, the *Encryption/Decryption* phase comes. All plaintext *P* or ciphertext *C* is padded into 64-bit data blocks and also processed with state *S* by 6 rounds permutation as the *Associated Data* phase do. The output ciphertext or plaintext is generated at this phase. The last phase *Finalization* contains 12 rounds permutation which is the same as the *Initialization* phase. The authentication tag *T* is generated at this stage to guarantee the correctness of the message source. If the data is forged or tampered by an attacker, authentication will fail at this stage and the decrypted plaintext will not be output.

ASCON takes an SPN-based permutation *P* as round transformation. It contains 12 iterative execution of *P* in the *Initialization* and *Finalization* phase and 6 in others. The permutation *P* consists of a constant-addition operation p_C, a substitution layer p_S and a linear diffusion layer p_L[5]. The 320-bit state *S* is split into five 64-bit register words x_0, x_1, x_2, x_3, x_4 where x_0 represents the Most Significant Bit (MSB). Each round *P* starts with the constant-addition operation p_C which adds a round constants c_r to the register word x_2 of the state *S*. The constant c_r changes with round *i* as $c_{r\mathrm{i}} = (\ 0\mathrm{xF})\ ||\ 0\mathrm{x}0 + \mathrm{i}$, where || means the concatenation of the two parts. After p_C, the substitution layer p_S is applied which consists of 64 parallel 5-bit S-boxes. The S-box is worked with each bit-slice of the five registers $x_0, \ldots, x_4$, where x_0 acts as the MSB and x_4 as the LSB of the S-box. It can be easily implemented in its bit-sliced form. As it is highly parallel, 320-bit *S* can be easily processed by 64 S-boxes simultaneously. The detail calculation of S-box sees appendices table 3. Finally, the linear diffusion layer provides a linear transformation to the S-box output. It contains five applications of the function to each 64-bit word of the register $x_0, \ldots, x_4$. See appendices (1) to (5) for the specific calculation formula.

3. DETAIL OF CHIP REALIZATION

3.1 Low area lightweight chip implementation

The block diagram and data path of our AE chip is depicted in figure 2.

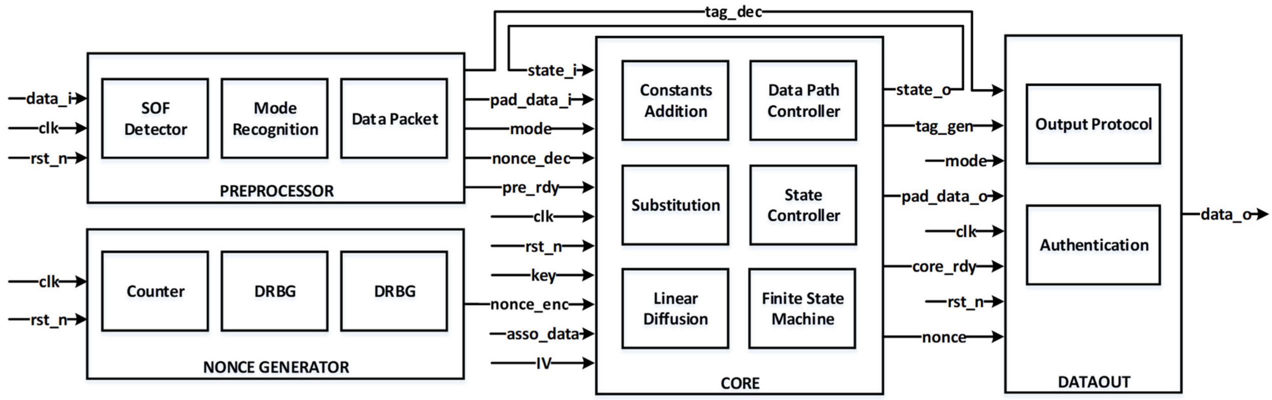


Figure 2. Block diagram of hardware implementation

The *PREPROCESSOR* and *DATAOUT* are responsible for data input and output processing. The former includes *SOF Detector*, *Mode Recognition* and *Data Packet* modules, which is mainly responsible for data reception, recognition and preparation. The latter consists of *Output Protocol* and *Authentication* modules.

The *NONCE GENERATOR* is a 128-bit pseudo-random nonce generator. It contains two 16-bit Deterministic Random Bit Generators (*DRBG*) and a 3-bit *Counter*. It is well known that the area of *DRBG* is proportional to the number of bits, so using two 16-bit *DRBG* to generate 128-bit pseudo-random *N* indirectly can save considerable hardware cost.

The *CORE* is the core algorithm which correspond to the four phases of ASCON. In order to reduce the area and improve the utilization of hardware, permutation function was selected as the main body, and the data was iterated through the round transformation itself. Each phase is switched by the master state machines *State Controller*, and the rounds within the state are controlled by nested slave state machines. The *Data Path Controller* controls all data paths and the state output is sent back to the input again to form its own loop. In this way, the algorithm core only needs a permutation function and peripheral digital selection logic to realize. The main body of the function runs at full load in the working state, which greatly improves the utilization efficiency of the hardware and reduces the area.

3.2 Mode of operation

When the valid data comes from RX, the *PREPROCESSOR* detects the *SOF* “1000” in the bit streams and starts information capturing, specifically, it stores the relevant data to the corresponding register. Since *Mode* is in the higher significant bit, the *PREPROCESSOR* can then decide whether to obtain the optional data frame nonce *N* and tag *T* according to the working mode, which can reduce both the amount of data transmission and the time of data processing.

The *CORE* consists of four phases of ASCON from *Initialization* to *Finalization*. The state *S* iterates itself according to the round transformation of the algorithm. At the same time, the associated data and plaintext (ciphertext) are grouped and absorbed into the *CORE* according to the principle of sponge function. After the ciphertext (plaintext) and tag *T* are generated, the *CORE* generates a $\mathrm{core\_rdy}$ signal to stop iteration and start the *DATAOUT* module.

The last part is *DATAOUT*, it performs differently according to the *Mode*. When the process is encryption, the *DATAOUT* combines *SOF*, *Mode*, ciphertext, nonce *N* used in the encryption, tag *T* generated by *Finalization* into a data stream and transmits it. When this process is decryption, the *DATAOUT* first judges whether the tag *T* obtained in the decryption process is consistent with the tag in the received data frame. If they match, the *DATAOUT* will output the decrypted plaintext only, otherwise no information will be output.

3.3 Low power design

Power consumption is a very important indicator[6]. For this work, the AE chip should have the ability to be used in resource or power constrained environments[7]. Therefore, power optimization is pretty important in the design process.

In order to reduce dynamic power consumption, many low power technologies were adopted in the design[8]. The first is the optimization of data path. The feature of ASCON is that the round function execution is accompanied by a large number of data selection logic, especially in the *Data Path Controller* module. An example of the data gating optimization in ASCON is shown in Figure 3. During the *Finalization* phase of ASCON, the x_1 of state *S* needs to be XORed with the higher 64 bits of key *K* before the first-round transformation in this stage. We know clearly that this operation is performed only at this round transformation, and that in most other cases x_1 is directly send to subsequent operations. Its specific principles are shown in the figure 3(a). In most cases, x_1 does not participate in XOR operation, but the XOR gates in the figure will always operate due to the continuous change of the input data, resulting in additional dynamic power consumption. In this example, the output of XOR gates is cut off at MUX. The optimized circuit is shown in figure 3(b). Operand Isolation (OI) is applied to the input of XOR. When the output of XOR is not selected, the selected signal of MUX cutoff the input of XOR through the and gate. This example illustrates the principle of this scheme, which is widely used elsewhere in this design for power optimization.

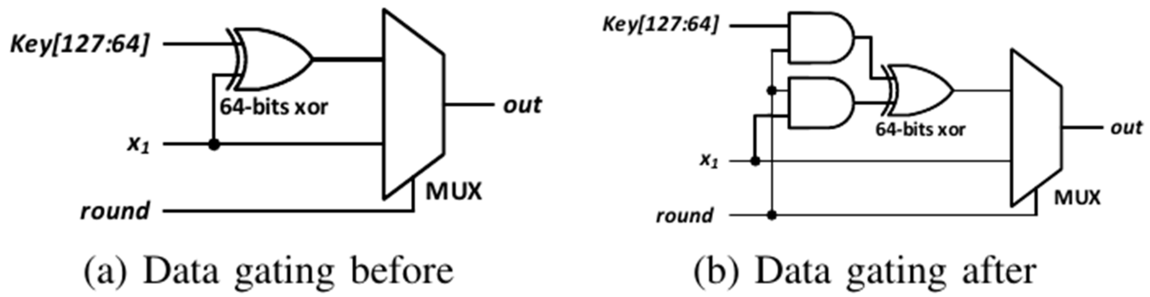
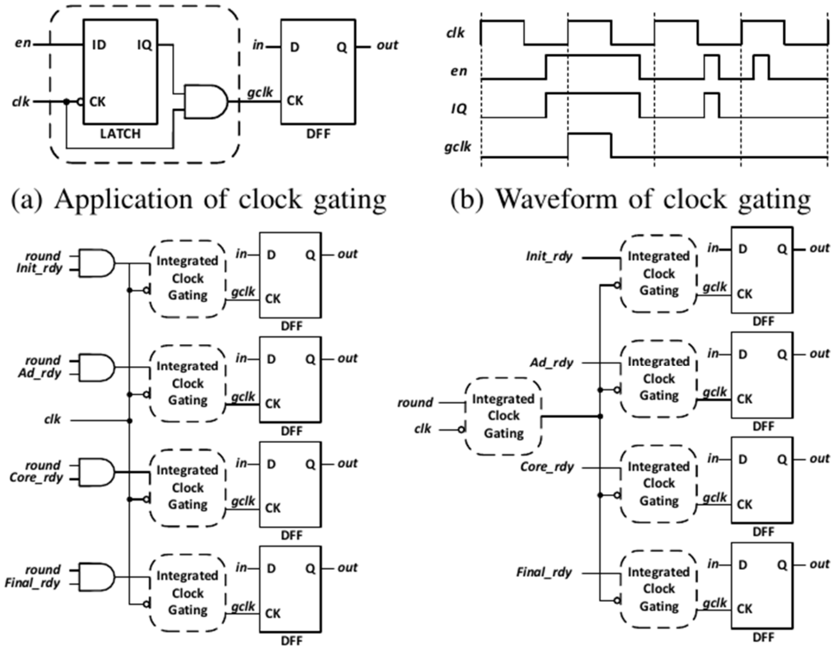


Figure 3. Data gating in this work

The next part is the optimization of clock path. Since the clock signal is very sensitive to glitches, turning off the clock directly through the Boolean logic gate will affect the signal quality. So, we used integrated clock-gating (ICG) latches to solve this problem. The ICG unit contains a negative level effective latch together with an AND gate. The *clk* is passed at positive *en* and closed at negative *en*. The glitches in *en* are filtered, whether it appears at the positive or negative level of clock as depicted in the waveform of figure 4(b). More than that, for some cases in our design, using ICG instead of enabling signal or data signal selecting logic can further optimize power consumption. An example in ASCON implementation is depicted in figure 4(c) to illustrate the detailed principle. In this case, the ICG units has already been added to clock path and the enabling signal was controlled by *round* and $*\_\mathrm{rdy}$ separately. We can modify this structure by use two stage of ICG units like figure 4(d) does. The first stage ICG unit selects & enable the next stage ICG units by gating their clock path to achieve the same function. In this situation, an ICG unit replaces four AND gates, resulting in 37.25% dynamic power reduction, 59.35% leakage power reduction and 25% area optimization.



(a) Application of clock gating (b) Waveform of clock gating

(c) Normal clock gating structure (d) multi-stage clock gating structure

Figure 4. Clock gating and other optimizations in this work

In addition, a series of general low-power methods were also used in our implementation. Such as low power placement, buffer insertion and gate-level optimizations. By using this series of low power designs, the PrimeTime PX (PTPX) reported total power of the chip is reduced from 23.1mW @125MHz to 15.4mW @125MHz, resulting in 33.3% optimization. Specifically, the internal power, switching power and leakage power are reduced by 32.6%, 33.2% and 99.9% (table 1). It should be noted that this work involves the application and optimization of low-power technology in ASCON design, rather than to invent a new low-power technology theory. Our goal is to provide a full range of work to guide the people who are interested in AE hardware research and chip-level implementations.

Table 1. Results of power optimizations

Power Classification (mW @125MHz)	Original design	Low-power design	Reduction
Internal power	15.6	10.5	32.6%
Switching power	7.418	4.949	33.2%
Leakage power	0.002843	0.0005781	99.9%
Total power	23.1	15.4	33.3%

4. RESULTS AND DISCUSSION

The whole AE chip for radio frequency application was fabricated under SMIC 180nm Mixed-Signal Ultra-High-Density process and packaged in SOP-28. Figure 5 shows the optical photo of the die and the packaged chip. The input data was generated by Xilinx Kintex-7 FPGA KC705 and the output waveform of the chip were characterized by DSLogic U3Pro16 logic analyser. Figure 5 depicted the output waveform of the chip and the partially enlarged view of both input and output under the global clock of 125MHz. Keithley 2410 1100v SourceMeter was used to measure the power consumption of the whole chip, which also serves as a power supply. The test result was 17.4mW at 125MHz @1.8V V_{DD}. Since the PTPX's power ratio shows that the core algorithm accounts 33.3% for the total power, we can roughly estimate that the power of core algorithm is about 5.8mW(46.4µW/MHz).

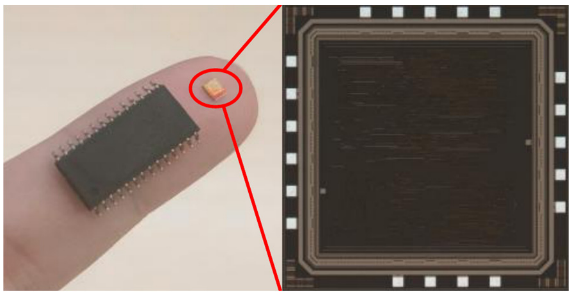


Figure 5. Optical photo of die and packaged chip

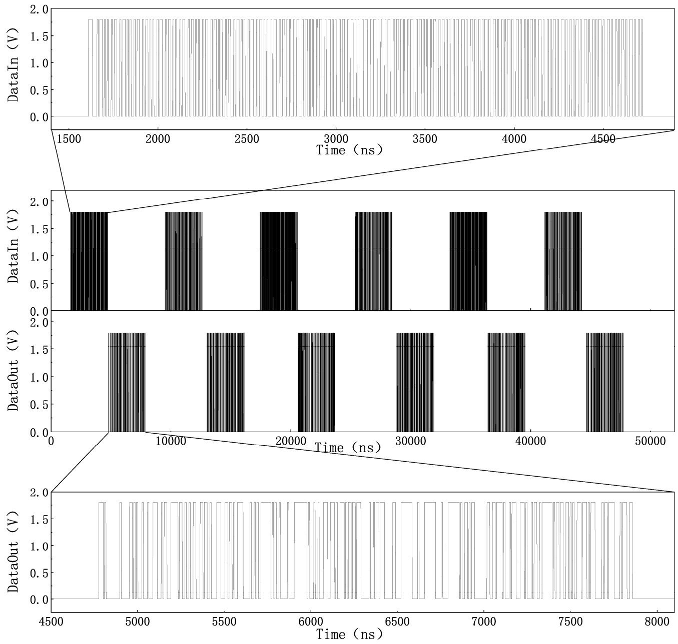


Figure 6. Test output of the whole chip

We compared our work with other relevant paper in table 2. Our proposed system is the only chip-level implementation. Despite the backward of technology and voltage, our chips still achieve a considerable chip area 0.461 mm^2, high throughput 1.33 bytes/cycle and low power consumption (46.4μW/MHz for core algorithm and 139.2μW/MHz for whole chip). Most importantly, it is the application of ASCON AE algorithm in the radio frequency field and can be applied independently in the radio frequency communication environment (which can also be used as an IP in a system).

Table 2. Comparison with other state-of-the-art

	This work	ASCON-fast[9]	ASCON-DOM[10]
Technology	SMIC 180nm	UMC 90nm	UMC 90nm
Voltage supply V	1.8	1	1
Area mm^2	0.461	NA	NA
Degree of completion	Chip-level	Back-end routing	Front-end synthesis
Core power μW/MHz	46.4	43	NA
Throughput bytes/cycle	4/3	4/3	4/9

5. CONCLUSIONS

An authenticated encryption chip for radio frequency application was developed which turned out to achieve low area, high throughput and low power dissipation. The whole chip achieved an ultra-high throughput of 1.07Gbps under 100MHz global clock with a reasonable chip area of 0.461mm^2 (1.4mm^2 with IOs) under SMIC 180nm technology. A series of low power method which turned out to achieve an ultra-low power dissipation of 46.4μW/MHz of core algorithm

(139.2μW/MHz with extra RF peripheral circuit and IOs) @1.8V V_{DD}. Therefore, it can be directly used as authentication encryption and decryption chip, but can also be used as IP embedded in other systems in a variety of occasions, such as the Internet of Things, etc.

6. APPENDICES

The following system of equations (1) to (5) shows the principle of linear diffusion layer of permutation. The calculations are made within each of the five 64-bit register word $x_0, \ldots, x_4$.

$$\sum\nolimits_0 (x_0) = x_0 \oplus (x_0 >>> 19) \oplus (x_0 >>> 28) \quad (1)$$

$$\sum\nolimits_1 (x_1) = x_1 \oplus (x_1 >>> 61) \oplus (x_1 >>> 39) \quad (2)$$

$$\sum\nolimits_2 (x_2) = x_2 \oplus (x_2 >>> 1) \oplus (x_2 >>> 6) \quad (3)$$

$$\sum\nolimits_3 (x_3) = x_3 \oplus (x_3 >>> 10) \oplus (x_3 >>> 17) \quad (4)$$

$$\sum\nolimits_4 (x_4) = x_4 \oplus (x_4 >>> 7) \oplus (x_4 >>> 41) \quad (5)$$

The following Table 3 shows the S-box input and output of permutation.

Table 3. The 5-bit S-box of permutation

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
S(x)	4	11	31	20	26	21	9	2	27	5	8	18	29	3	6	28
x	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
S(x)	30	19	7	14	0	13	17	24	16	12	1	25	22	10	15	23

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